



Historical Perspective

Hydrophilic polymer-based anti-biofouling coatings: Preparation, mechanism, and durability

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ABSTRACT

Anti-biofouling materials that combat microorganism attachment have been intensively studied due to the ever-growing demand on smart and durable coatings. Although various hydrophilic polymer surfaces demonstrated superior anti-biofouling properties, their practical application was hampered by the undesired mechanical vulnerability and complicated fabrication process. In this review, we summarized the mechanically and chemically robust anti-biofouling coatings into six strategies namely (i) 3D-grafted coatings, (ii) hierarchical spheres-based coatings, (iii) inorganic nanomaterials-reinforced coatings, (iv) hydrolysis-based coating, (v) semi-interpenetrating structure-based coatings, and (vi) layer-by-layer (LbL) assembled coatings. The anti-biofouling efficacy and durability of these coatings over a series of challenges were also comprehensively presented. The purpose of this review is to inspire researchers to develop novel anti-biofouling coatings for future practical applications.

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Contents

1. Introduction	1
2. Durability characterization methodologies	2
3. Biofouling characterization methodologies	2
4. Strategies to develop durable coatings	3
4.1. 3D-grafted coatings	3
4.2. Hierarchical spheres-based coatings	5
4.3. Inorganic nanoparticles-reinforced coatings.	6
4.4. Hydrolysis-based coatings	9
4.5. Layer-by-layer assembled surfaces	10
4.6. Semi-interpenetrating network-based coatings	10
4.7. Other durable anti-biofouling surfaces/coatings	11
5. Conclusions	14
6. Future outlook	14
Acknowledgments	14
References	14

1. Introduction

Bacterial attachment and colonization are ubiquitous phenomena which jeopardize a wide spectrum of areas ranging from healthcare to industries, including biomedical equipment, implantable devices, food

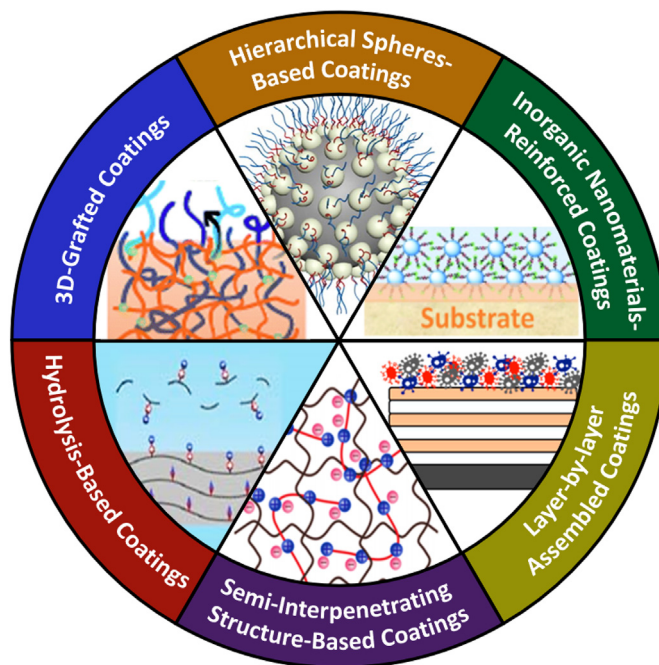
processing pipelines, and marine vehicles, etc. The Center for Disease Control and Prevention (CDC) reported that approximately 1.7 million patients in the US acquired healthcare-associated infections (HAIs) annually and over 98,000 of them deceased due to the infection [1]. The massive additional costs related to HAIs for patient families and healthcare systems range from 28.4 to 33.8 billion USD each year [2]. The Agency for Health care Research and Quality pointed out that the HAIs had already become one of the top ten causes of death in the US

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[3]. According to World Health Organization (WHO), around 30% of patients in intensive care units (ICU) are affected by at least one HAI even in high-income countries [4]. Studies also indicate that healthcare facilities can act as amplifiers for the wide spread of life-threatening infections including SARS, Ebola, etc. [5]. One major cause for the HAIs is the environmental surfaces in healthcare facilities are contaminated by frequent touch from patients and healthcare workers [6,7]. To mitigate the financial burden and unnecessary deaths posed by HAIs, significant research activities have been directed at preparing antibacterial coatings to protect the high-touch surfaces in healthcare facilities. In the past decades, various chemical strategies were developed to impart antimicrobial coatings with active-killing properties which include release-killing (e.g. nanoparticles and/or ions of silver, copper, zinc, selenium, quaternary ammonium compound, or antibiotics) [8–18], contact-killing (e.g. peptides or quaternary ammonium polymers) [19–27], and light-activated killing (e.g. photosensitizers) [28–30]. However, the preloaded antimicrobial agents will exhaust upon long-term use and the dead microbial debris caused by active-killing coatings could serve as a protective layer for the approaching microorganism to colonize and grow in a “live-on-dead” fashion [31–33]. Moreover, the overuse of antibiotics also led to the emergence of multi-drug resistant (MDR) bacteria [34–36].

To address those problems, a wide variety of hydrophobic polymer-based coatings were investigated to obtain desired anti-biofouling property. Despite the fact that they have demonstrated excellent mechanical durability and water-proofness, limited anti-fouling performance was achieved due to the hydrophobic interaction and polar attraction between the coatings and the foulants [37]. In recent years, surface hydrophilization was broadly utilized as a new paradigm to eliminate microorganism colonization. Surface hydration layer induced by hydrophilic polymers can confer anti-biofouling property to surfaces since a layer of tightly bound water functions as an energetic and physical barrier which effectively resists biofouling process, including protein attachment, initial bacterial attachment, and subsequent biofilm formation [38–43]. Several hydrophilic polymers were employed as candidates to develop hydrophilic anti-biofouling coatings. Among them, poly(ethylene glycol) (PEG), zwitterion-containing polymers and their derivatives were widely investigated since they can form hydration layers via hydrogen bonding and electrostatic attraction [44–49]. Owing to their water solubility and poor mechanical durability, hydrophilic polymers cannot be solely applied as coatings [50]. Enormous strategies were adopted to improve the durability of the hydrophilic polymers which can be summarized into two subcategories: (i) surface functionalization of hydrophobic surfaces, e.g. construct hydrophilic polymer brushes via atom transfer radical polymerization (ATRP) or photo initiation [51–53], (ii) copolymerization or physical mixing with hydrophobic polymers, e.g. methyl methacrylate (MMA) [54–56], polyurethane (PU) [57–59], or polydimethylsiloxane (PDMS) [60–62]. But none of these techniques demonstrated potential practical applications so far. Specifically, the construction of polymer brushes usually requires complicated chemistries and inert atmosphere and the limited thickness of the brush layer rendered the coatings susceptible to mechanical damage. On the other hand, the content of PEG or zwitterions can be incorporated into hydrophobic moieties by simple copolymerization or physical blending is limited, higher amount of hydrophilic polymers will certainly sacrifice overall mechanical durability of the coatings [60,63,64]. Hence, novel strategies and techniques were extensively investigated to impart desired properties to the hydrophilic polymers. In this review, we will comprehensively overview the recent development of the anti-biofouling coatings based on hydrophilic polymers with an emphasis on mechanical durability. Recent advances on the design and development of durable anti-biofouling coatings were summarized based on their working mechanisms namely (i) 3D-grafted coatings [65–70], (ii) hierarchical spheres-based coatings [71–73], (iii) inorganic nanomaterials-reinforced coatings [74–80], (iv) hydrolysis-based coatings [50,81–85], (v) semi-interpenetrating structure-based coatings [86–88], and (vi) layer-by-layer (LbL) assembled coatings [89–95] were summarized in Scheme 1.



Scheme 1. Strategies to prepare mechanically and chemically durable hydrophilic polymer-based coatings.

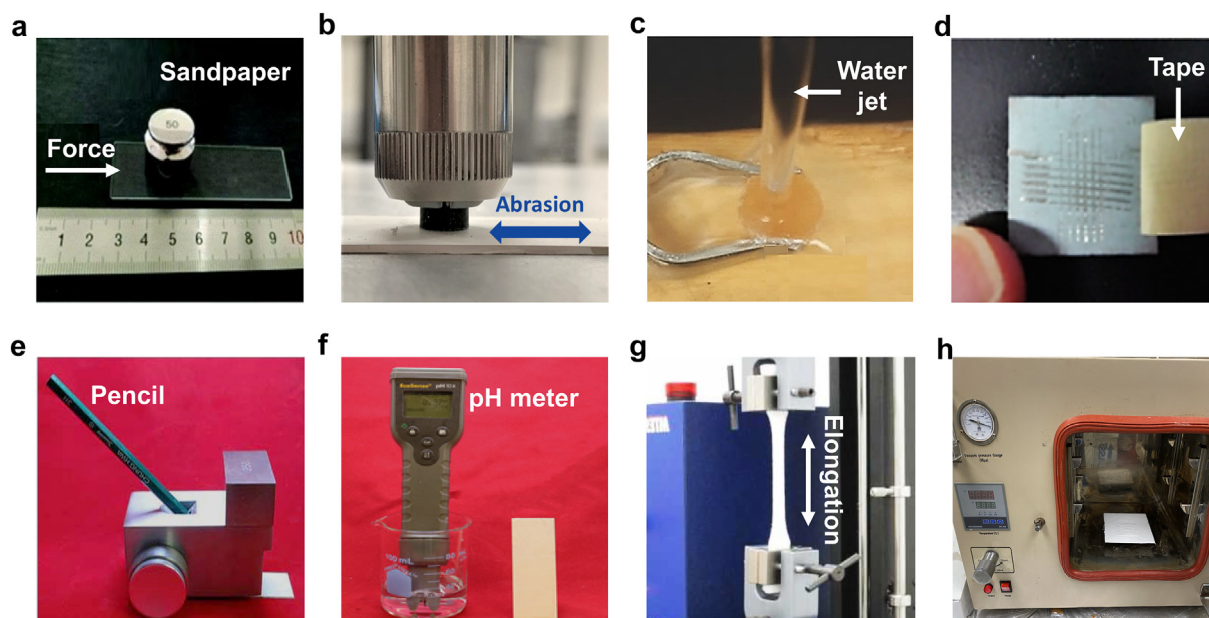
2. Durability characterization methodologies

The evaluation of the durability of anti-biofouling coatings is mainly focused on its robustness against mechanical, chemical, and environmental challenges. To confirm the feasibility of these coatings for practical applications, a plethora of methodologies have been derived based on ASTM or ISO standards. Scheme 2 shows the photographs of the most commonly applied techniques to measure the durability of the anti-biofouling coatings, including sandpaper abrasion test (Scheme 2a), Taber abrasion test (Scheme 2b), water jetting/flushing test (Scheme 2c), adhesion tape test (Scheme 2d), pencil hardness test (Scheme 2e), solution immersion test (Scheme 2f), tensile strength test (Scheme 2g), and thermal stability test (Scheme 2h). A more comprehensive summary with typical characterization procedure and related standards is provided in Table 1. Given the fact that most research groups conducted the same durability test based on different non-standard self-adjusted procedures, care must be taken when comparing the results from different research groups.

3. Biofouling characterization methodologies

The anti-biofouling properties of the bactericidal-free hydrophilic polymer-based coatings are usually characterized by evaluating the resistance toward the attachment of proteins and/or microorganisms and the formation of biofilm. For the protein test, bovine serum albumin (BSA) and bovine fibrinogen (BFG) are commonly used as model proteins. As for microorganism test, Gram-positive bacteria (e.g. *Staphylococcus aureus*, *Staphylococcus epidermidis*, and *Bacillus subtilis*) and Gram-negative bacteria (e.g. *Escherichia coli*, *Pseudomonas aeruginosa*), algae (e.g. *Amphora coffeiformis*), diatom (e.g. *Navicula incerta*), and cells are frequently utilized.

The observation and quantification of the attached microorganisms and biofilms can be generally accomplished by numerating viable cells with plate count method or flow cytometry. Microscopy techniques ranging from simple light microscopy to the state-of-the-art confocal laser scanning microscopy (CLSM) can also be used for cell counting and biofilm 3D characterization. Surface morphology and topography



Scheme 2. Photographs of (a) sandpaper abrasion test, (b) Taber reciprocating abrasion test, (c) water jetting/flushing test, (d) adhesion tape peeling test, (e) pencil hardness test, (f) chemical resistance test, (g) tensile strength test, and (h) thermal stability test.

of the biofilms can be determined using atomic force microscopy (AFM), scanning electron microscopy (SEM). Moreover, surface plasmon resonance (SPR), quartz crystal microbalance with dissipation monitoring (QCM-D) are adopted to investigate biofouling phenomenon on a molecular scale. The commonly used qualitative and quantitative biofouling characterization methodologies regarding their advantages and figures of merit are summarized in Table 2.

4. Strategies to develop durable coatings

4.1. 3D-grafted coatings

Grafting hydrophilic polymers onto mechanically and chemically robust hydrophobic coatings is a broadly applied technique to obtain desired surface hydrophilicity and anti-biofouling properties. Conventional grafting route exhibited limited durability due to the detachment and degradation of the hydrophilic polymers, leading to failure for long-term service. To overcome the drawbacks of traditional approach, Minko et al. proposed a novel 3D grafting method to obtain

prolonged durability by grafting hydrophilic polymers to both the surface and inside the matrix [65]. Specifically, poly(ethylene oxide) (PEO) was grafted into cross-linked poly(2-vinyl pyridine) (P2VP) network via reaction between the alkyl-halogen groups in PEO and P2VP pyridine rings (Fig. 1a). When the PEO brush on the surface was damaged, the underlying PEO segments stored in the matrix will spontaneously emerge to the surface due to the gradient in chemical potential. The 3D-grafted surface thus demonstrated longer durability by four-fold in physiological condition compared to the conventional regime. Similarly, Zhou et al. reported a facile UV-curing route to achieve long-lasting anti-biofouling property by grafting poly(ethylene glycol) dimethacrylate (PEGDA) into the matrix of a UV-curable epoxy resin (BEA) for underwater applications [66]. As shown in Fig. 1b, the PEGDA segments were casted onto solid-like BEA sheet and allowed to inter-diffuse for a certain time. Afterward, the excess PEGDA was poured off and the epoxy coating was cured under UV irradiation. The 3D-grafted layer showed prolonged underwater stability with a thickness remained nearly constant for up to 70 days. However, this 3D-grafted epoxy coating exhibited limited long-term

Table 1
Durability characterization methods and standards.

Test	Description	Standard	Ref.
Sandpaper abrasion test	A coated-substrate is placed face-down to the sandpaper with a weight attached to the substrate. The substrate is pushed back and forth.	N/A	N/A
Taber abrasion test	The coating film is abraded by rotating the panel under weighted abrasive wheels.	ASTM D4060 ISO 7784-2	[96,97]
Water jetting/flushing test	Coating film is subject to water jetting/flushing for a specified flow rate and time, followed by examination for deterioration.	N/A	N/A
Tape peeling test	The adhesion of the coating to the substrates is evaluated by applying and peeling Scotch tapes over cross-cut region.	ASTM D3359 ISO 2409	[98,99]
Pencil hardness test	The hardness of the coating film is determined by pencil leads of known hardness ranging from 6B to 6H.	ASTM D3363 ISO 15184	[100,101]
Solution immersion test	Chemical resistance of the coating film is measured by immersing in solutions (e.g. DI or salty water, acidic or alkali solutions, and organic solvents, etc.) for a specified time, followed by examination for deterioration.	ASTM D870 ASTM D1308	[102,103]
Tensile strength test	Free-standing coating film with specified dimension is subject to tensile elongation by tensile testing apparatus, the tensile strength, elongation, and modulus can be thus determined.	ASTM D2370	[104]
Thermal stability test	The coating film is placed in an oven at elevated temperature for a specific time, followed by cooling to ambient temperature and examination for deterioration.	ISO 3248	[105]

Table 2
Biofouling characterization methodologies.

	Metrology	Figures of merit	Notes
Quantitative	Confocal laser scanning microscopy (CLSM)	Biomass ($\mu\text{m}^3/\mu\text{m}^2$) Biovolume (μm^3) Thickness (μm) Coverage (%)	High resolution 3D imaging, quantitative assessment of biomass and thickness of the biofilms
	Plate count	Colony forming units/mL (cfu/mL)	Only live cells will be enumerated; Well-standardized protocol
	Quartz crystal microbalance – dissipation (QCM-D)	Frequency shift (Δf) dissipation shift (ΔD)	Real-time assessment of the biofilm formation; Calibration between frequency and cell count is required
	Scanning Electron Microscopy (SEM)	N/A	High resolution and magnification
Qualitative	Atomic force microscopy (AFM)	Surface roughness (R_q)	High resolution 3D imaging of the topography

anti-biofouling durability due to the hydrolysis and degradation of PEG segments.

Alternatively, zwitterionic polymers as non-fouling materials with improved stability can also be employed for 3D grafting. Wu et al.

employed a novel zwitterionic triblock copolymer for constructing 3D-grafted self-repairing polyurethane (PU) coatings [67]. Briefly, a copolymer with building blocks of poly(ethylene glycol) (PEG), poly(2-hydroxyethyl methacrylate) (PHEMA), and poly(2-

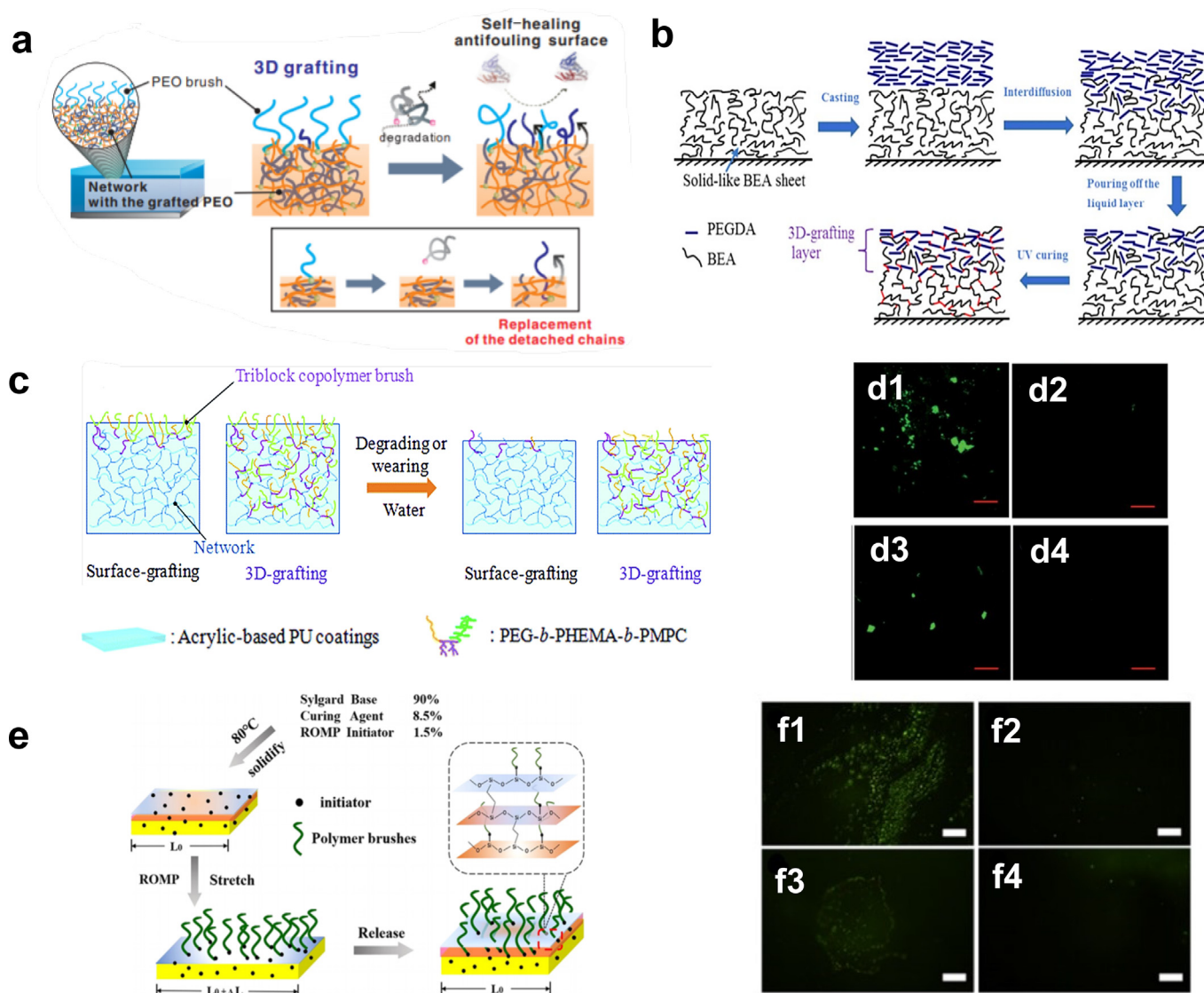


Fig. 1. (a) Schematic illustration for 3D grafting of PEO into P2VP polymer matrix and the self-healing process after PEO segments were detached from surface. Reprinted with permission [63]. Copyright 2013, John Wiley & Sons. (b) Preparation procedure of the PEGDA 3D-grafted UV-curing epoxy coatings. Reprinted with permission [64]. Copyright 2016, John Wiley & Sons. (c) Mechanism of the self-repairing of the zwitterionic triblock copolymer 3D-grafted PU coating after degrading or wearing. (d) CLSM images of FITC-BSA attachment on (d1) surface grafted coating after immersion in PBS buffer for 30 days; (d2) 3D-grafted coating after immersion in PBS buffer for 30 days; (d3) surface-grafted coating after polishing; (d4) 3D-grafted coating after. The scale bars are 100 μm . Reprinted with permission [65]. Copyright 2015, Royal Society of Chemistry. (e) Preparation of zwitterionic polymer brush modified PDMS via subsurface-initiated ROMP. (f) Fluorescence microscope images of *S. aureus* (f1-2) and *E. coli* (f3-4) bacterial attachment on (f1, 3) PDMS and (f2, 4) robust thick zwitterionic polymer brush grafted PDMS. Scale bars are 20 μm . Reprinted with permission [68]. Copyright 2019, American Chemical Society.

methacryloyloxyethyl phosphorylcholine) (PMPC) was spray-coated onto pre-cured acrylic-based PU coatings via the reaction between hydroxyl groups in the copolymer and the isocyanate groups in PU (Fig. 1c). Compared with the previous work, this PU-based 3D-coating can be heat-cured without any UV irradiation process and the PMPC segments potentially have better stability than PEG. The stability of the coating was evaluated by immersing in PBS buffer for 30 days. As a result, the 3D-grafted coating showed nearly constant hydrophilicity while the WCA of conventional surface grafted coating was substantially increased. The durability of the 3D-grafted coating is owing to the self-replenishing of the hydrophilic copolymers from inside to the surfaces. This 3D-coating also demonstrated superior long-lasting antifouling property against BSA attachment even after 30 days of immersion. Fig. 1d shows the CLSM images of the FITC-BSA protein adsorption on the coatings, as for the long-term durability in PBS, the 3D-grafted coating (d2) displayed dramatically decreased fluorescence intensity compared with surface grafted coating (d1) after immersion in PBS for 30 days. Similarly, reduced brightness was also observed from mechanically polished 3D coating (d4). These observations indicated the 3D grafted coating can effectively maintain anti-biofouling property over long-term service.

Besides grafting the as-synthesized zwitterionic polymers into matrix via “graft-to” strategy to construct 3D coatings, “graft-from” technique which develops the polymer brushes from the subsurface area of the matrix was also exploited. In 2019, Zhou et al. developed robust thick zwitterionic polymer brushes based on poly(5-norbornene-2-ylcarboxyethyl-1-imidazolium propane sulfonate) (PNCIPS) via a novel subsurface-initiated ring-opening metathesis polymerization (SSI-ROMP) technique [70]. Unlike conventional surface-initiated atom transfer radical polymerization (SI-ATRP) grafting, the 3D grafting of zwitterionic polymer brushes was initiated from ROMP initiators embedded in the subsurface area of the PDMS elastomers (Fig. 1e). Due to the rapid kinetics of ROMP technique, the thickness of the zwitterionic polymer brushes can be up to microns which endows desired durability to the as-prepared surfaces. The antibacterial attachment performance

was confirmed using fluorescence microscopy (Fig. 1f), brilliant fluorescence signal from *S. aureus* and *E. coli* colonization was diminished to nearly black after the PDMS elastomers were grafted with robust thick zwitterionic polymer brushes.

4.2. Hierarchical spheres-based coatings

Underwater superoleophobicity and anti-biofouling properties are the keys to marine non-fouling coatings. Wu et al. developed a brand-new self-healing underwater coating via the assembly of hierarchical microgel spheres [71]. This coating demonstrated superior durability upon challenged with physical, chemical, and environmental harsh conditions. The synthetic procedure for the modified hierarchical microgel spheres (MHMS) was depicted in Fig. 2a, N-isopropylacrylamide (NIPAM), methacrylic acid (MAA), and poly(ethylene glycol) diacrylate (PEGDA) were first used to fabricate the microgel spheres (MS), followed by the addition of silica nanoparticles to obtain hierarchical microgel spheres (HMS). Then, poly(methacryloxy propyl trimethoxysilane)-*block*-(poly(hydroxyethyl methyl acrylate)-*co*-poly(2-methacryloyloxyethyl phosphorylcholine)) (PMPS-*b*-(PHEMA-*co*-PMPC)) hydrophilic block copolymers were grafted to the silica nanoparticles on HMS to achieve modified hierarchical microgel spheres (MHMS). The MHMS was subsequently spin-coated on glass substrate and allowed for self-assembly, followed by spin-coating of thermoset acrylic polyurethane resin onto MHMS layer to get the durable anti-biofouling coating. This coating can retain its underwater superoleophobicity even when challenged with highly acidic or alkali environments (Fig. 2b). Upon mechanical abrasion, the coating surface can easily recover its original oil-repellent property by immersing in water for 12 h (Fig. 2c1, c2). When being crushed, the surface can also restore its nanostructure after immersing in water for 25 min (Fig. 2c3, c4). The remarkable self-healing property was attributed to the swelling of MHMS and the replenishment of hydrophilic copolymers to surface when submerged in water. Fig. 2d shows the CLSM images of FITC-BSA protein attachment on the (d1)

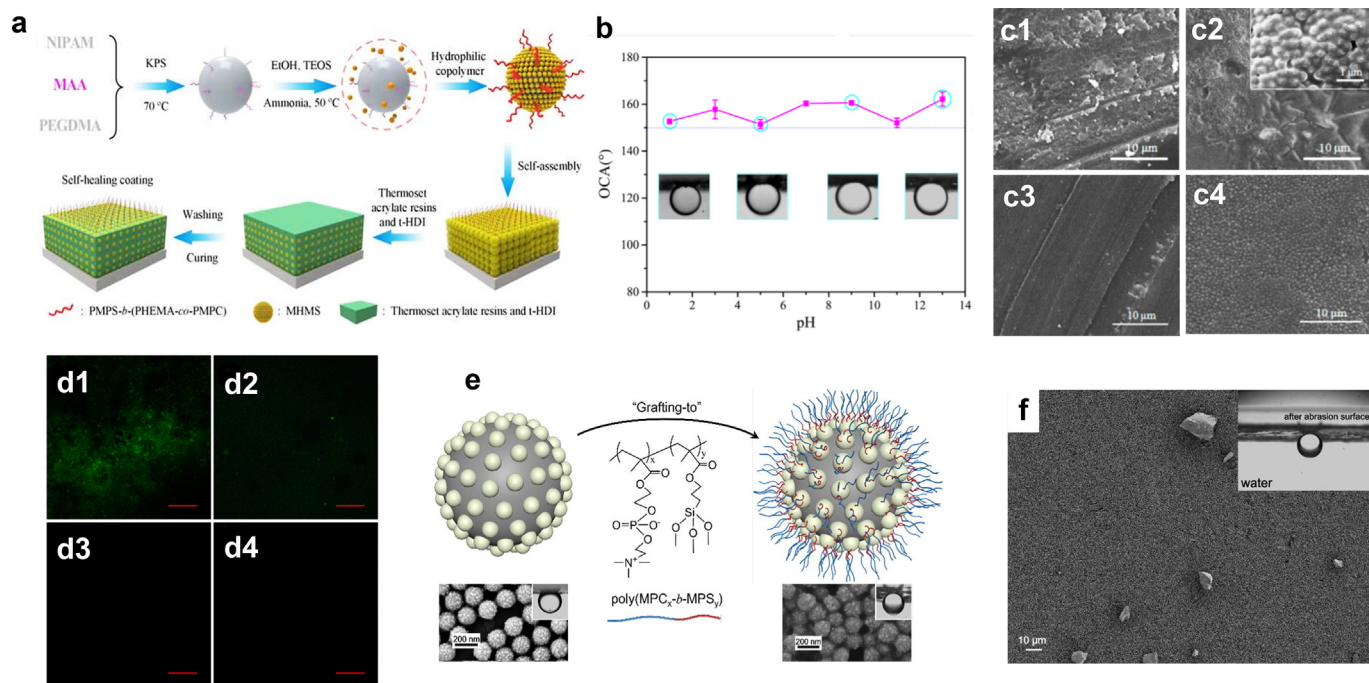


Fig. 2. (a) Preparation of the modified hierarchical microgel sphere (MHMS)-based coatings. (b) Underwater OCAs of MHMS-based coatings in acidic or alkali solutions. (c1) and (c2) MHMS-modified coatings after mechanical abrasion and recovered after immersing in water; (c3) and (c4) Crushed and restored coating surfaces. CLSM images of (d1) unmodified PU coating; (d2) original MHMS-modified coating; (d3) abrasion-repaired MHMS coating; (d4) long-term incubated MHMS coating. Reprinted with permission [69]. Copyright 2016, American Chemical Society. (g) Schematic illustration of grafting hydrophilic block copolymer onto raspberry-like polymer/SiO₂ hybrid colloidal spheres. (f) Surface morphology and OCA of the hybrid sphere-based coating after sand abrasion. Reprinted with permission [71]. Copyright 2018, Elsevier.

PU coating, (d2) MHMS-modified coating, (d3) abrasion-repaired MHMS-coating, and (d4) PBS buffer 30 days incubated coating. As a result, almost no fluorescence signal was observed on the MHMS-modified coating. Even when the coating is repaired from mechanical damage or exposed to long-term incubation, the MHMS coating can still efficiently resist protein adsorption, indicating the excellent anti-biofouling and self-repairing performance.

In 2018, Wu and co-workers presented a simplified route to prepare hierarchical spheres-based coating without any PU binder [73]. In brief, raspberry-like hybrid colloidal spheres were prepared by grafting zwitterionic block copolymer poly(MPC-b-MPS) onto poly(ethyl methacrylate)/SiO₂ spheres (Fig. 2e). The coating films were formed by siloxane condensation between the copolymers thus no binder was required. Sand abrasion test was used to confirm the anti-abrasion property of the coating film. As shown in Fig. 2f, no distinct exfoliation or detachment were observed from the coating surface which is attributed to the outstanding film forming property and the interaction between the hybrid colloidal spheres and glass substrates. Moreover, the biofouling-resistant property was measured with BSA attachment assay, a minimum value of $0.44 \mu\text{g cm}^{-2}$ was obtained from this research which is smaller than that of $3 \mu\text{g cm}^{-2}$ in the previous work.

Since the hydration layers are prone to be breached over long-term service, a combination of bacteria-repellent, release-killing, and contact-killing properties are preferred for constructing a robust and versatile surface to eliminate bacterial colonization. Guo et al. designed

a new anti-biofilm coating based on modified polystyrene (MPS)/Ag microspheres with combined bacteria-repellent and bactericidal abilities [72]. The structure of the multifunctional copolymer and the hierarchical microspheres are illustrated in Fig. 3a, the lotus-like MPS/Ag microsphere structure was obtained by grafting a copolymer of poly(ethylene glycol maleate) (PEGMA), glycidyl methacrylate (GMA)–iminodiacetic acid (IDA) conjugate (GMA–IDA), and QAS segments onto the hydroxylated polystyrene microspheres (HPS). The long-lasting bactericidal property was achieved by the synergistic effect of the release-killing silver nanoparticles (Ag NPs) chelated on GMA–IDA moieties and the contact-killing QAS segments. Fig. 3b displays the fluorescence microscopy images of the live/dead assay for both *S. aureus* and *E. coli* bacteria on bare substrate and MPS/Ag microspheres cast films. Brilliant red fluorescence images were observed from both types of bacteria, suggesting excellent bactericidal performance of the microspheres. On the other hand, the MPS/Ag microspheres can be easily mixed with epoxy curing agent to prepare hydrophilic-type coating with superior mechanical durability. The mechanical durability was demonstrated by abrasion and impact test, correspondingly, no exfoliated debris were observed on the coating surface (Fig. 3c, d).

4.3. Inorganic nanoparticles-reinforced coatings

Hydrophilic polymers grafted nanoparticle as an easily modifiable and highly versatile platform was extensively investigated for a wide

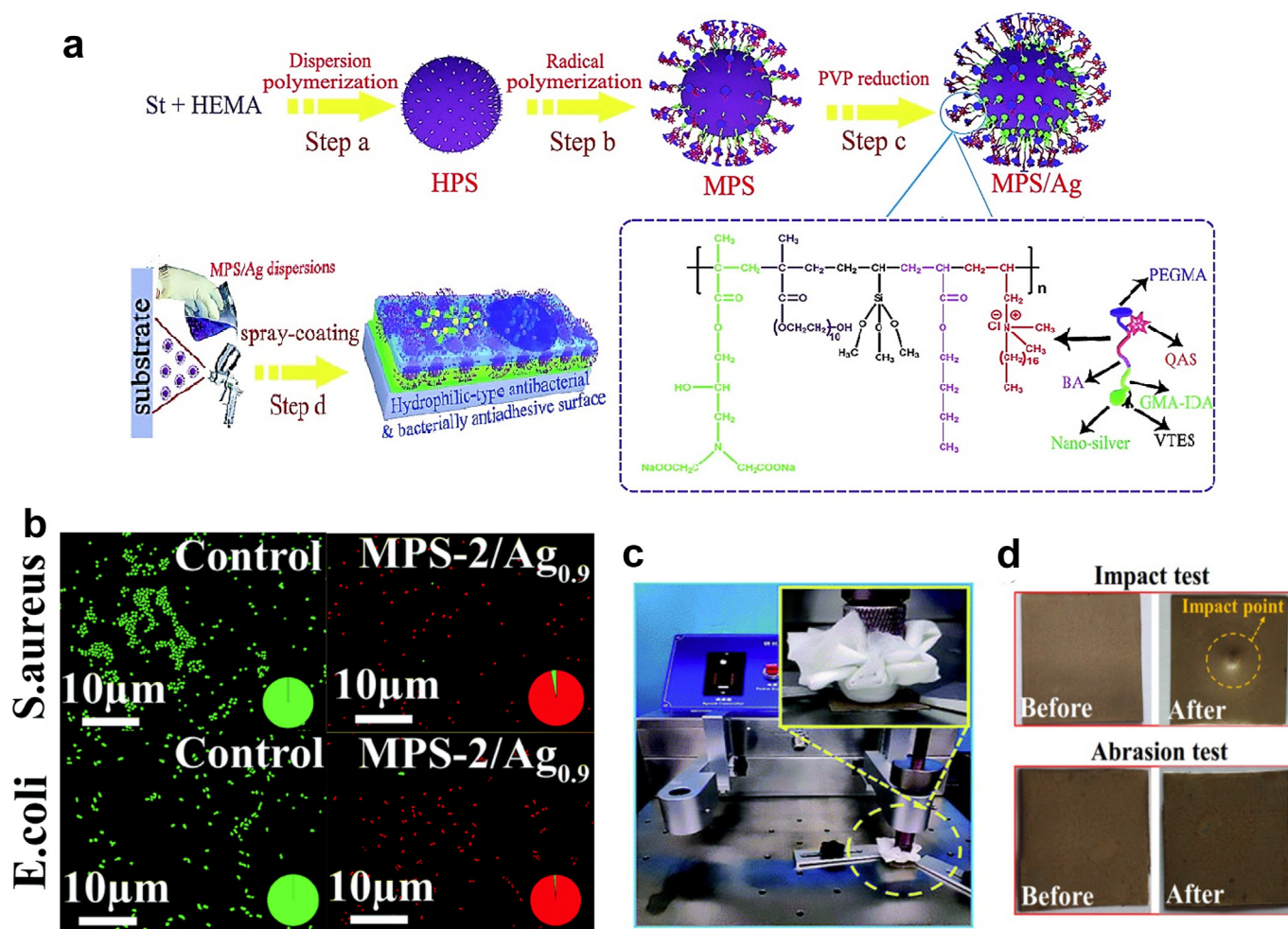


Fig. 3. (a) Schematic for the preparation of the hydrophilic-type coating based on MPS/Ag microspheres. (b) Fluorescence microscopy images of live/dead assay for both *S. aureus* and *E. coli* bacteria adhered to bare substrates or MPS/Ag microsphere coatings. (c) Photograph of the mechanical abrasion test setup. (d) Photographs of the hydrophilic-type coatings before and after impact and abrasion tests. Reprinted with permission [70]. Copyright 2019, Royal Society of Chemistry.

range of applications. To solve the intrinsic drawbacks of poor mechanical durability of the hydrophilic polymers, the structure of a polymer shell around inorganic nanomaterials was designed to obtain combined durability and surface hydrophilicity. He and co-workers successfully implemented this idea to fabricate a mechanically robust underwater superoleophobic coating with ameliorated anti-corrosion and anti-biofouling properties [74]. A dual-layered structure was prepared by casting methacryloxy propyl trimethoxysilane (MPS)-SiO₂/PNIPAM (N-isopropyl acrylamide) hybrid nanoparticles (HNS) onto epoxy resin (ER) binder interlayer to obtain the superoleophobic and anti-biofouling coating (Fig. 4a). The ER interlayer served as physical support for the hybrid nanoparticles and the hierarchical structure of these nanoparticles. The as-prepared HN/ER coatings exhibited underwater OCAs of above 150° after 24 h immersion which confirmed the tolerance toward harsh chemical environments (i.e. strong acidic, alkali, and ionic strength solutions) (Fig. 4b). Moreover, this coating can also withstand 40 cycles of sandpaper abrasion without significant change in BSA protein adsorption, compared with ER coating, the abraded HN/ER coating

showed negligible increase in the fluorescence intensity which exemplified its robustness against mechanical wear (Fig. 4c).

Similarly, Wang et al. also developed a durable marine anti-biofouling coating using sequential deposition method. In their approach, zwitterionic polymers was functionalized onto silica nanoparticles to achieve a structure of poly(sulfobetaine methacrylate)/vinyltrimethoxysilane/silica (PSBMA@VTMO@silica) and a mixture of tannic acid (TA) and poly(ethylene glycol) (PEG) was employed as binder layer to attach the nanoparticles onto substrates [75]. Fig. 4d shows the adhesion test on coating surface with cross-cut. As a result, only less than 5% of area was detached after 3 times of peeling which indicates a superior adhesion grade of 4B. The abrasion resistance test was performed by abrading the coating using a sandpaper loaded with a 50 g weight. As shown in Fig. 4e, the surface retained its hydrophilicity after 12 cycles of abrasion. The long-lasting hydrophilicity can be attributed to two main causes: (i) the physical support of TA/PEG interlayer and the mechanical robustness of inorganic silica nanoparticles; (ii) the hierarchical structure formed by stacking of the hybrid nanoparticles which endowed prolonged hydrophilicity even when the top layer

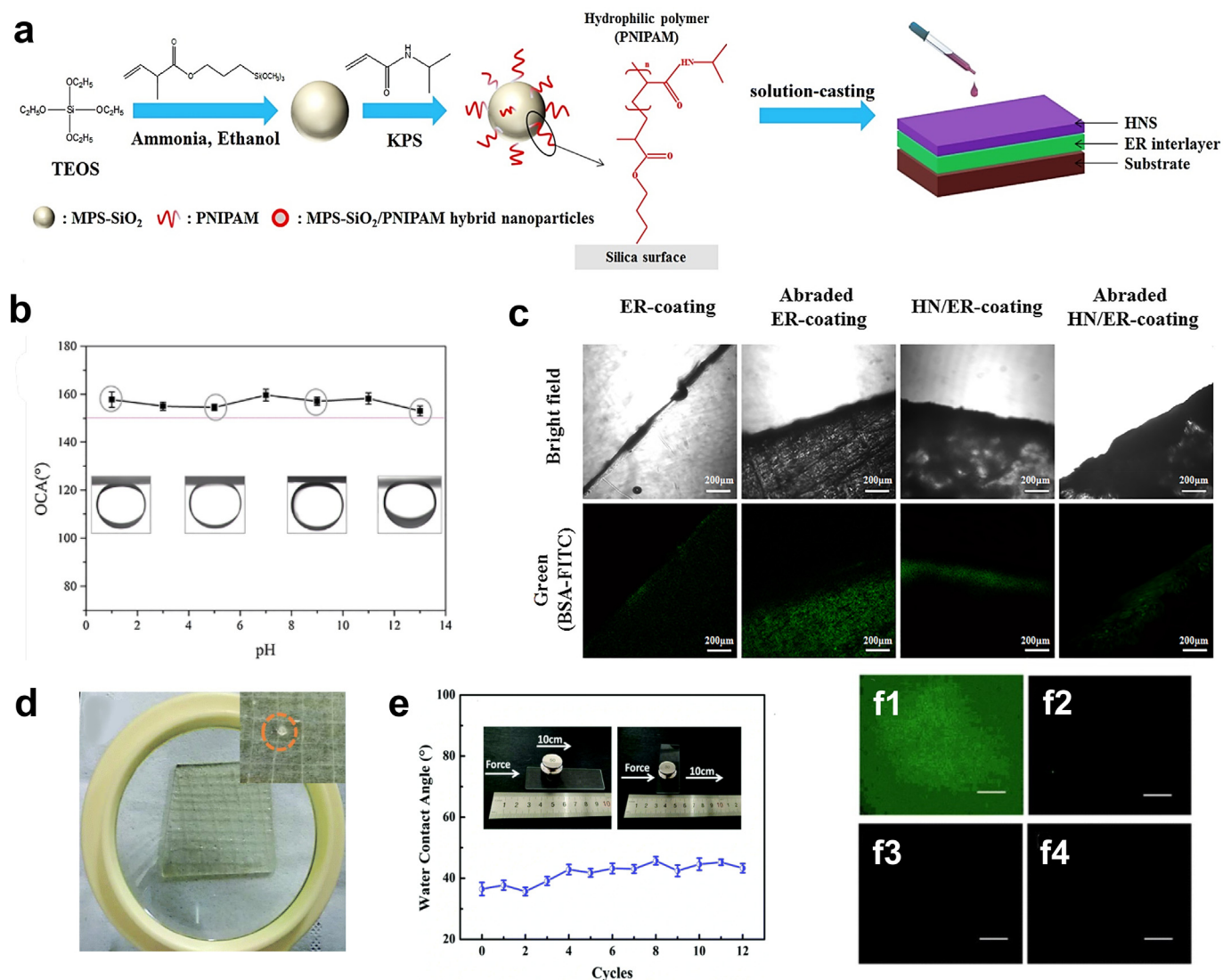


Fig. 4. (a) Synthetic route and preparation of the durable underwater superoleophobic coating. (b) Underwater OCAs of the superoleophobic coating after immersing in acidic and alkali solutions. (c) Bright field and fluorescence microscopy images of BSA-FITC adsorption on original and abraded ER and HN/ER coatings. Reprinted with permission [72]. Copyright 2018, Elsevier. (d) Photographs of the nanocomposite coating after tape peeling test. (e) WCA evolution after 12 cycles of sandpaper abrasion and photographs of the abrasion test setup. (f) Fluorescence images of BSA attachment on (f1) TA-PEG coating, (f2) ~ (f4) original, abraded, and, peeling tested nanocomposite coatings. Scale bars are 200 μm. Reprinted with permission [73]. Copyright 2020, Royal Society of Chemistry.

was removed by mechanical damage. Moreover, this nanocomposite coating can also endure a high temperature of more than 150 °C, the thermogravimetric analysis showed an onset of mass loss at 197 °C. Fig. 4f displays the fluorescence images of BSA attachment on TA-PEG coating (f1), original nanocomposite coating (f2), abraded (f3), and peeling tested (f4) coatings, respectively. Surprisingly, all of the tested samples are capable of retaining almost “zero” BSA protein adsorption. These results further validated the outstanding long-lasting anti-biofouling property of this nanocomposite coating.

PEG as another commonly used hydrophilic polymer can also be grafted onto silica nanoparticles. Zhou and co-workers developed SiO₂-PEG based acrylic polyurethane (APU) coatings by simple blending for long-lasting outdoor dirt resistance application [76]. After exposed

to outdoor environment for 42 days, the nanocomposite coatings can still maintain a relatively clean surface whereas the control samples were heavily contaminated with dirt.

Even the abovementioned coatings can tolerate a series of the mechanical challenges, severe mechanical damage such as thick blade cut and high cycles of abrasion can still potentially incur failure for long-term service. Hence, self-healing ability of the nanomaterial reinforced coatings is highly demanded. Su et al. fabricated scratch-resistant anti-fogging coatings with rapid self-healing property based on hydrophilic copolymers of poly(sulfobetaine methacrylate) and poly(2-hydroxyethyl methacrylate) (p(SBMA₇-co-HEMA₃)) and silica nanoparticles grafted with sulfobetaine zwitterionic moieties [77]. As illustrated in Fig. 5a, the damaged coating can self-heal with the assistance of water

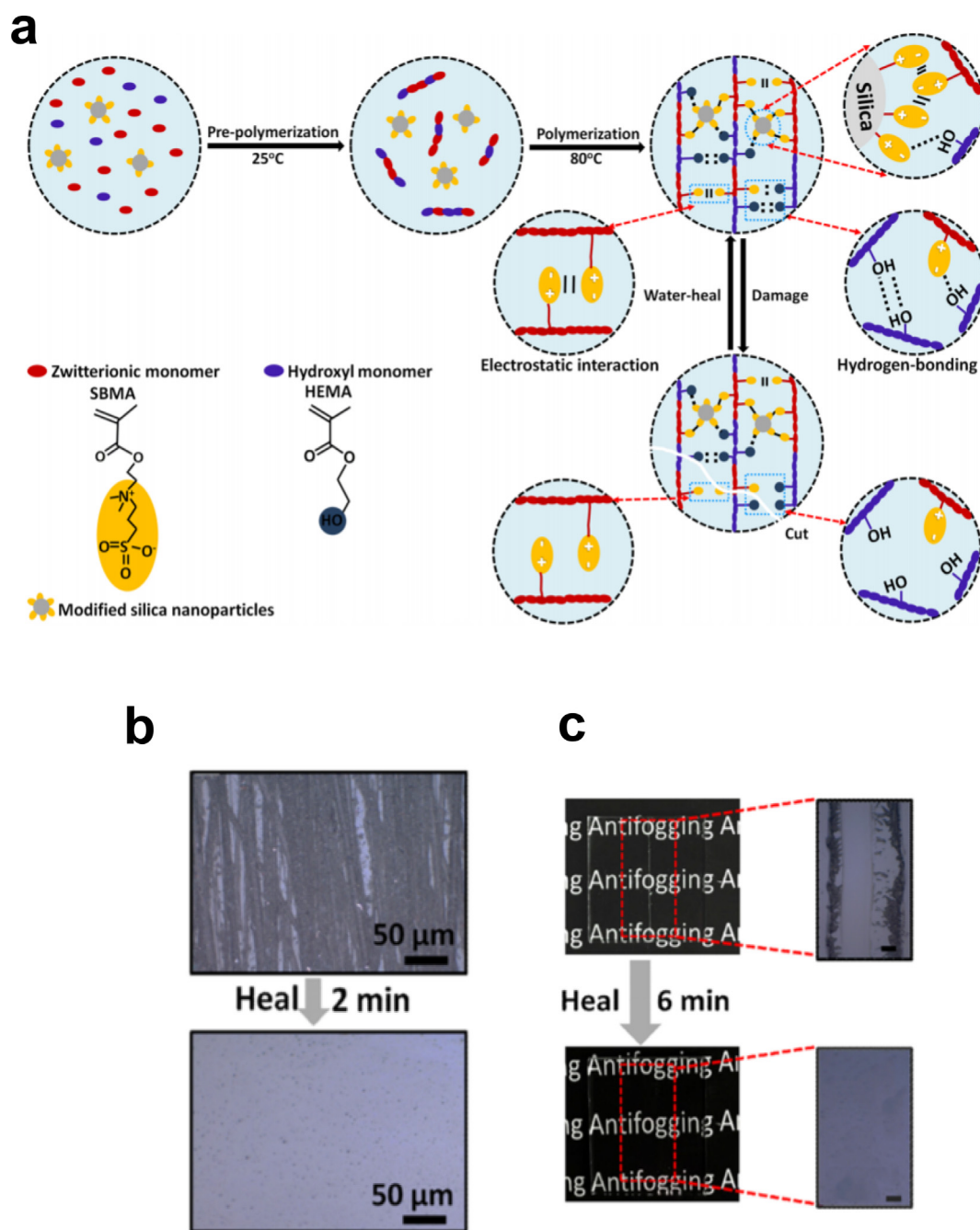


Fig. 5. (a) Preparation of the durable self-healing anti-fogging coating. (b) Optical images and photographs of the self-repairing process for mechanically abraded coatings. (c) Optical images and photographs of the self-healing process for coating with blade cut. Reprinted with permission [75]. Copyright 2019, American Chemical Society.

through the reversible electrostatic attraction and hydrogen bonding between the hydrophilic copolymers and the zwitterion-functionalized silica nanoparticles. Upon abraded with sandpaper, the grooves generated by scratch can be readily fixed by dipping into DI water for only 2 s and the composite coating will heal in ambient environment for 2 min (Fig. 5b). Most importantly, even when the coating was severely damaged, it still can be fixed by the same process. Fig. 5c displays a cut made by a blade with a thickness of 250 μm and subsequently restored by immersing in water. In addition, the biofouling resistance of this coating is also investigated with QCM-D. BSA protein attachment can be reduced by two thirds for the coated Au electrode.

4.4. Hydrolysis-based coatings

The regeneration of anti-biofouling property after mechanical damage is crucial to achieve long-term durability. Zhang and co-workers designed a self-renewing coating which regenerates zwitterionic moieties in seawater upon mechanical damage based on a copolymer of tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA), mechanically durable methyl methacrylate (MMA), and degradable 2-methylene-1,3-dioxepane (MDO) (Fig. 6a) [83]. The TCBSA monomer is able to hydrolyze into zwitterionic moieties in marine environments which renders the coating surface with persistent anti-biofouling property. However, the real marine environments are complicated thus the coatings solely based on TCBSA monomer are still inclined to be covered with inorganic foulants, leading to the failure of the self-generating ability. To address this problem, a degradable MDO monomer with ester linkage was introduced into the copolymer to obtain synergistic self-renewing properties. As for the mechanical durability, the coating

maintained an adhesion strength of over 2.0 MPa since the acrylic segments can form strong polar and hydrophobic interactions to the substrates. This coating can also retain a pencil hardness of HB over 28 days of immersion in seawater. The anti-biofouling ability was evaluated by investigating the adsorption of marine bacteria *Pseudomonas* sp., Fig. 6b shows the fluorescence microscopy images of both PMMA (b1, b3) and coating (b2, b4) surfaces without (b1, b2) and with pre-hydrolysis for 3 days (b3, b4). The PMMA surface was heavily covered with bacteria, while only a small group of bacteria can attach to the self-renewing coating surface, indicating TCBSA has successfully hydrolyzed into zwitterionic moieties even only exposed to aqueous for limited term.

Later in 2020, the same research group presented another eco-friendly hydrolyzable anti-biofouling polyurethane coating with poly (lactic acid) segments and natural antifoulants [84]. Compared with the previous work, bio-sourced poly(lactic acid) was utilized to build hydrolyzable segments in polyurethane for self-renewing property. In addition, a natural anti-biofoulant butenolide was also blended into the coating for synergistic anti-biofouling properties (Fig. 6c). To confirm the feasibility of practical application, long-term durability of the coating was tested in real marine environment for 3 months. The control PVC coating was heavily covered by marine organisms while only 7% of this novel polyurethane coating was covered with biofoulants.

Recently, Chen et al. reported a versatile way to prepare hydrolysis-based coating for combating marine biofouling [85]. In their approach, a series of monomers with functions of producing zwitterionic moieties upon hydrolyzation, cross-linking with poly(lactic acid), and increasing mechanical durability were copolymerized to combine desired properties. The working mechanism is depicted in Fig. 6d. Results showing

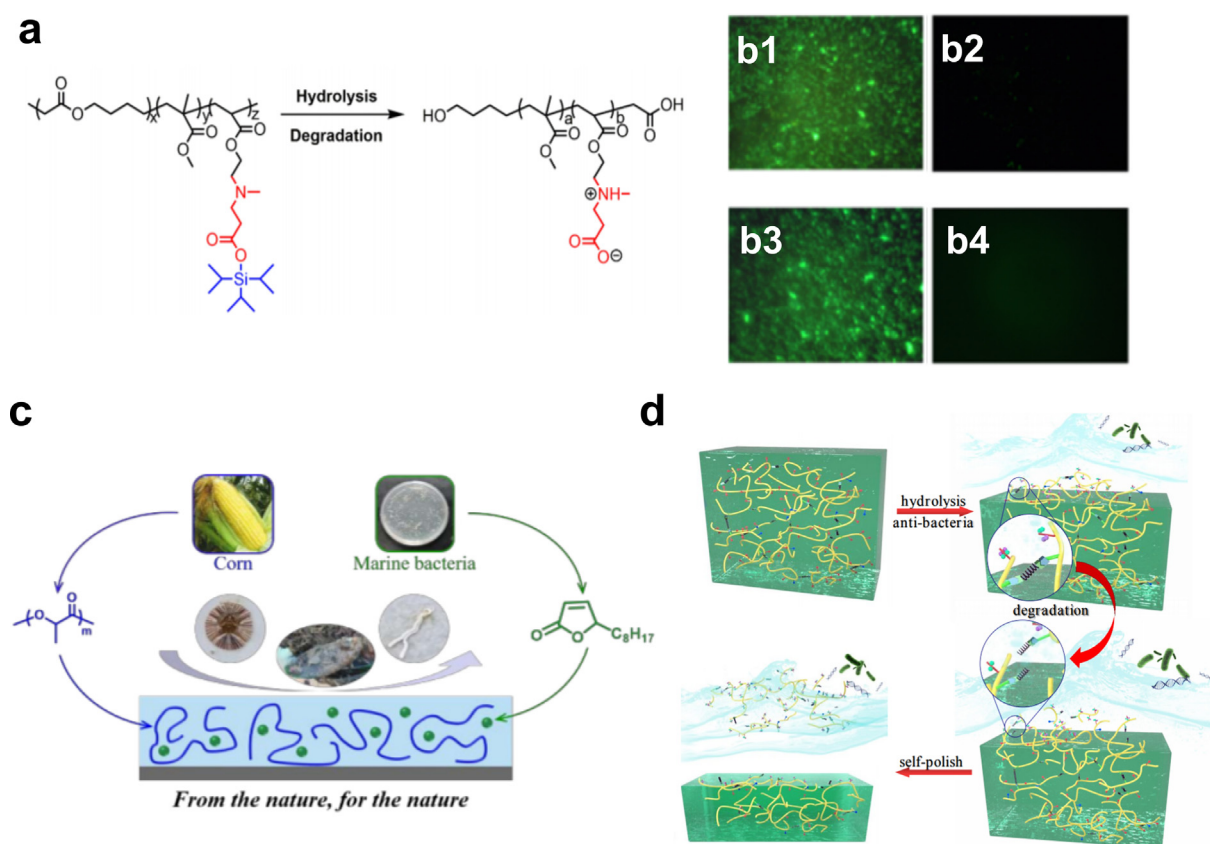


Fig. 6. (a) Chemical structure of the coating polymer after hydrolysis and degradation. (b) Fluorescence microscopy images of *Pseudomonas* sp. attached to PMMA (b1,b3) and coating surface (b2,b4) with (b1,b2) and without (b3,b4) pre-hydrolysis. Reprinted with permission [81]. Copyright 2019, American Chemical Society. (c) Schematic illustration of the bio-sourced self-generating and self-renewing coating. Reprinted with permission [82]. Copyright 2019, American Chemical Society. (d) Schematic of the long-term resistance to protein adsorption and bacterial biofilm formation through controllable zwitterionic surface regeneration. Reprinted with permission [83]. Copyright 2020, American Chemical Society.

that this coating can effectively prevent biofilm formation from both Gram-positive and Gram-negative bacteria for over 14 and 6 days.

4.5. Layer-by-layer assembled surfaces

Layer-by-layer (LbL) assembly for fabricating multi-layered thin films has received growing attention due to its simplicity and precise control over surface composition and thickness. Dong et al. employed this technique to prepare zwitterionic polymer surfaces on various substrates (including stainless steel, textiles, and wood, etc.) for durable anti-corrosion and anti-biofouling applications [89]. Fig. 7a illustrates the preparation of the coating, a polydopamine (PDA) layer was first deposited on the substrates to improve anti-corrosion ability and serve as binder for anchoring the successive layers. (3-aminopropyl) triethoxysilane (APTES) and poly(sulfobetaine methacrylate-co-glycidyl methacrylate) (PSB) zwitterionic copolymer were subsequently immobilized onto the substrates. The long-term durability of the assembled layers was comprehensively evaluated by measuring tensile strength after mechanical abrasion and immersion in harsh chemical environments. Maximum tensile strength of cotton textiles and wood plates upon immersion in 40 wt% H_2SO_4 solutions for 6 days was retained after coated with the multilayered coating. The robustness of the resistance toward protein attachment was assessed by measuring the change of BSA adsorption after 40 days of immersion. Surprisingly, the anti-biofouling remained almost constant which further corroborated the feasibility for long-term application.

PEG films can also be readily deposited onto various substrates via polydopamine mediated binding. Kang et al. designed a multilayered marine anti-fouling surface based on polydopamine and PEG catechol [94]. As shown in Fig. 7b, solid substrates were first spin-coated with polydopamine film, followed by the deposition of PEG catechol (PEG-C) layer. And these two layers were further cross-linked via catechol- Fe^{3+} -catechol interactions for optimized stability. The chemical resistance of the as-prepared films was determined by acidic or alkali

solution treatment, the thickness of PEG-C/ Fe^{3+} film was slightly decreased. Fig. 7c displays the fluorescence images of the diatom attachment assay for 1,2,3, and 4 weeks of immersion in seawater. The anti-biofouling potency was not deteriorated as only slight increase in attached diatom density was observed.

Compared with other common methods to fabricate multilayered films with long-lasting durability, layer-by-layer assembly demonstrated its desired simplicity to enable higher thickness by alternatively deposit the component layers. In 2020, Yang and co-workers have produced a freestanding self-healing coating for wound regeneration based on the interaction of cyclodextrin-modified silk fibroin (SCD) and adamantane-modified hyaluronic acid (HAD) (Fig. 7d) [95]. The self-healing ability was demonstrated by scratching the coating with a 30 μm thick scalpel then immersing in water. It was found that the coating was repaired by the host-guest interaction between SCD and HAD. This coating was also confirmed with the ability to restore repeated scratches at the same spot (Fig. 7e). On the other hand, this coating can inhibit the growth of *E. coli* by ~95% which is attributed to the anti-bacterial property of hyaluronic acid.

4.6. Semi-interpenetrating network-based coatings

The semi-interpenetrating polymer networks (SIPNs) consisting of one or more networks with inter-penetrated linear polymers were investigated for anti-biofouling and water purification applications. The SIPNs offered an easily tunable structure to obtain enhanced mechanical and chemical stability for hydrophilic polymers. Zhou et al. prepared SIPNs based on charged polymers and poly(vinyl alcohol) (PVA) for marine antifouling application [86]. As shown in Fig. 8a, sulfobetaine methacrylate (SBMA) and acrylamide (Am) monomers were copolymerized with cross-linker *N,N'*-methylenebisacrylamide (MBA) to form SIPN with PVA via in situ free radical polymerization. The mechanical properties of the charged SIPN was determined with tensile strength test, after incorporated into PVA networks, both tensile strength and elastic

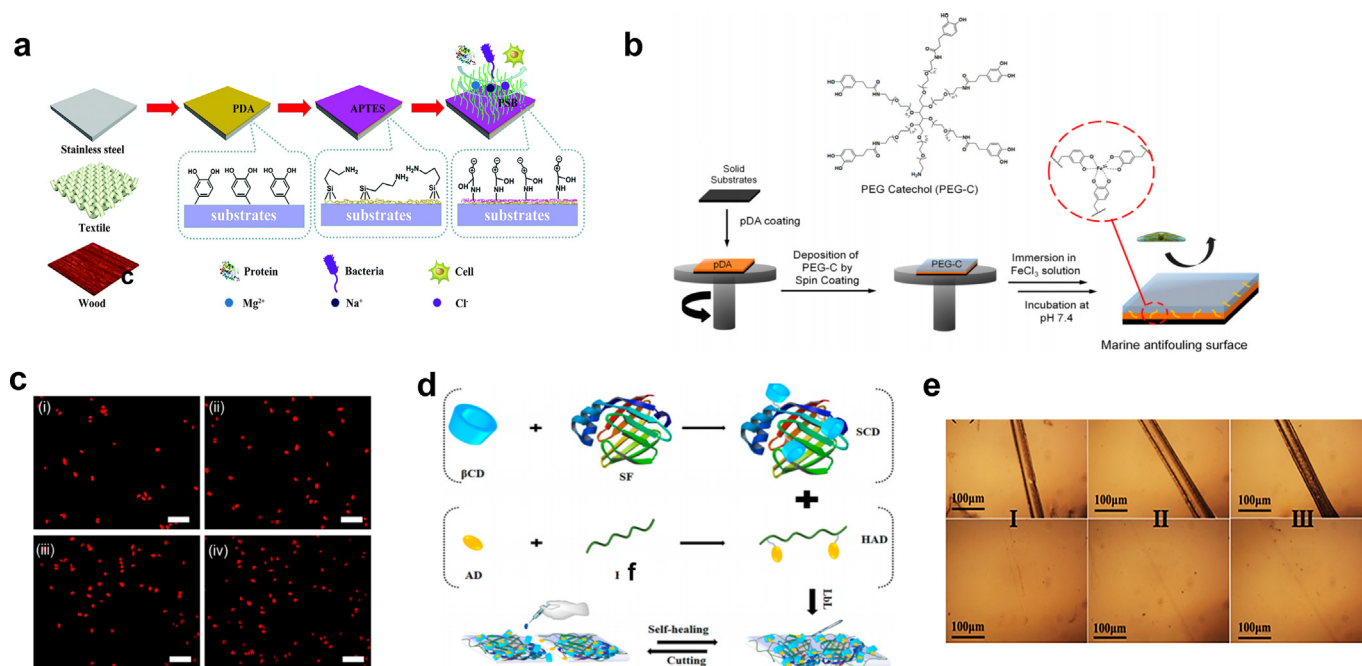


Fig. 7. (a) Schematic illustration of the layer-by-layer assembled zwitterionic coating surface. Reprinted with permission [87]. Copyright 2019, Royal Society of Chemistry. (b) Preparation of the multilayered PEG films on PDA-coated substrates and chemical structure of the PEG catechol (PEG-C). (c) Fluorescence images of diatom attachment assay after (i) 1, (ii) 2, (iii) 3, and (iv) 4 weeks of immersion in seawater. Scale bars are 50 μm . Reprinted with permission [92]. Copyright 2017, American Chemical Society. (d) Schematic illustration of the preparation of HAD/SCD coatings via layer-by-layer assembly. (e) Optical microscopy images of the self-healing process upon repeated scratch. Reprinted with permission [93]. Copyright 2020, American Chemical Society.

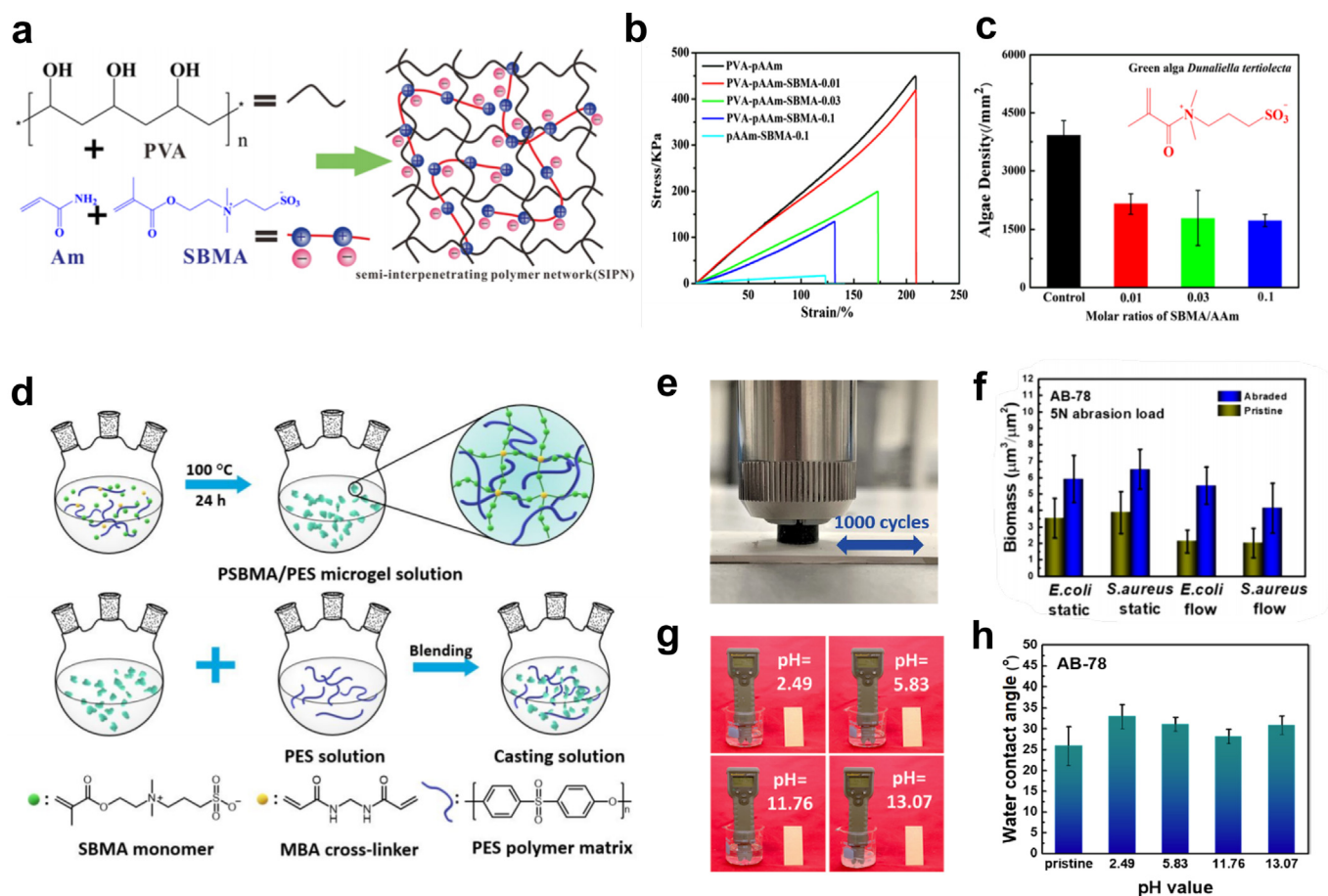


Fig. 8. (a) Schematic illustration of the SIPN of PVA and networked Am and SBMA hydrophilic copolymers. (b) Tensile strength of the SIPN coatings. (c) Algae densities on the zwitterionic SIPN coatings. Reprinted with permission [84]. Copyright 2017, American Chemical Society. (d) Synthetic route for the zwitterionic SIPN microgels and preparation of the coating solutions. (e) and (f) Photograph of the abrasion test setup and biomass of the attached bacteria on the abraded coating surfaces. (g) and (h) Chemical stability test for the coatings in various pH conditions and WCAs after immersed for 24 h. Reprinted with permission [85]. Copyright 2020, American Chemical Society.

modulus were substantially improved (Fig. 8b), indicating the mechanical property of the coating was reinforced by PVA. The marine anti-biofouling performance was assessed by green algae *Dunaliella tertiolecta* attachment assay, in Fig. 8c, the algae density on coatings with 0.01 M ratio of SBMA/AAm was reduced by 46% compared to the control PVA-pAAm sample.

The mechanical durability of SIPN coatings can be further improved by constructing with hydrophobic polymers. Poly(ether sulfone) (PES) as a widely used polymer for water purification membranes can be potentially utilized as a platform for anti-biofouling coatings due to its excellent mechanical durability, film-forming capability and chemical stability. Ghasemi et al. developed a novel mechanically and chemically robust anti-biofouling surface by converging the mechanical property of PES and the bacteria-repellent capability of zwitterionic poly(sulfobetaine methacrylate) (PSBMA) polymer using an SIPN microgel structure [87]. The synthetic route was depicted in Fig. 8d, the SIPN microgels were first synthesized to allow PES polymer chains to entangle with zwitterionic microgels. Then, the as-prepared microgels were further blended with PES casting solution for enhanced mechanical properties. Semi-interpenetrated zwitterionic microgels were thus successfully integrated with hydrophobic PES segments via the entanglement of the PES chains in microgels with the further blended PES chains in the casting solution. The long-term durability of the coating was determined by abrading the coating surface with a Taber abrader with a load of 5 N for 1000 cycles, the increment of biomass of *E. coli* and *S. aureus* bacteria adhered on the coating surfaces was still satisfactory. The repressed bacterial attachment indicated that the zwitterionic

microgels were not only loaded on the surface but also inside the coatings (Fig. 8e, f). Chemical stability in harsh chemical conditions (i.e. acidic and alkali solutions) as another key factor to achieve long-lasting durability was also determined by immersing in solutions with various pH values for 24 h. Surface hydrophilicity was retained nearly constant even when the coatings were challenged with solutions with pH values of 2.49 or 13.07 (Fig. 8g, h).

In addition, the enrichment of hydrophilic polymers on the surface of SIPN coatings is crucial for constructing robust hydration layers and ameliorated bacterial inhibitory property. Zhao and co-workers proposed a novel in situ post-crosslinking strategy to achieve the SIPN structure with higher surface content of zwitterionic moieties, the post-crosslinking process was fulfilled by the hydrolysis of methoxy groups in the 3-methacryloxypropyl trimethoxysilane (MTSi) silane coupling agent of the copolymer [88]. The as-prepared SIPN membranes demonstrated outstanding water purification efficacy and suppressed BSA adsorption. Also, this membrane can tolerate a temperature of ~300 °C, which showed its potential to be used for hot water filtration.

4.7. Other durable anti-biofouling surfaces/coatings

Medical device-related high-touch surfaces are often considered as the media for bacterial infection. To circumvent this problem, Sae-ung et al. prepared a universal anti-biofouling coating based on a copolymer of zwitterionic monomer methacryloyloxyethyl phosphorylcholine (MPC) and cross-linkable methacrylate-substituted dihydrolipoic acid

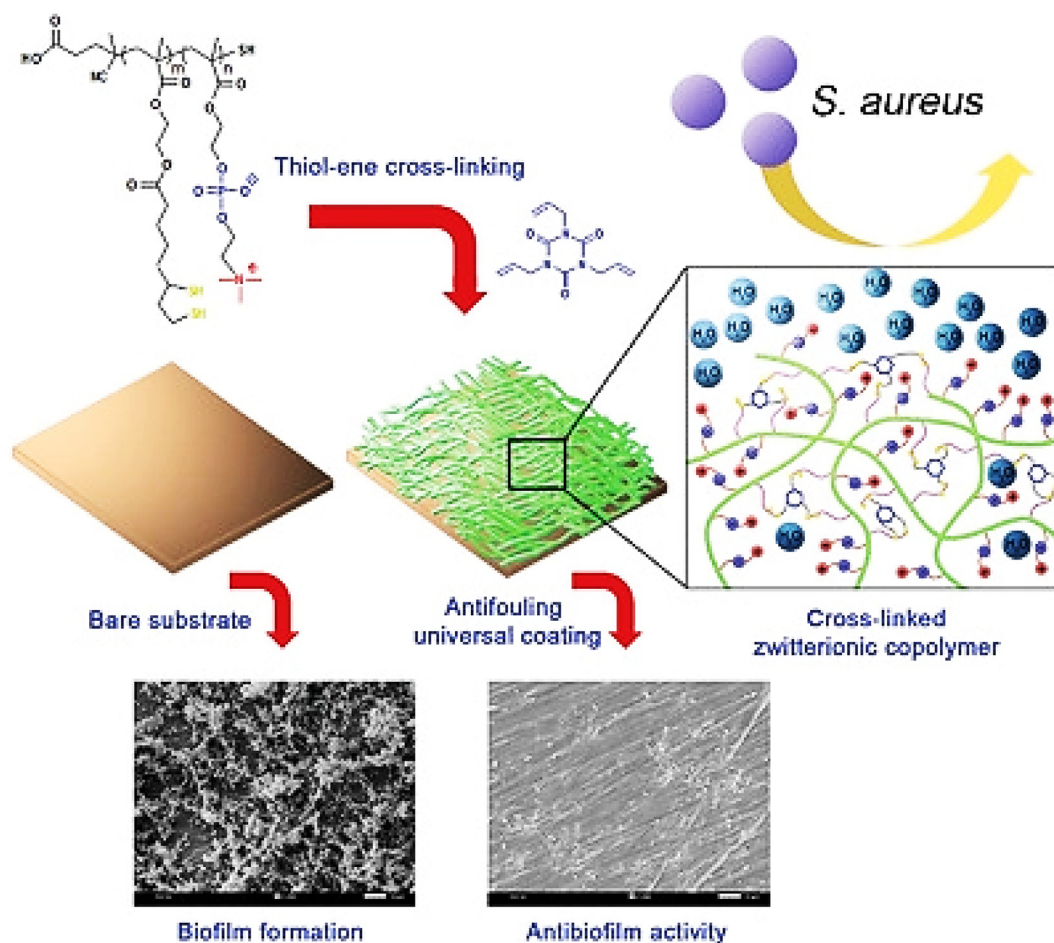


Fig. 9. Schematic illustration of the clickable zwitterionic copolymer as a durable anti-biofilm coating. Reprinted with permission [104]. Copyright 2019, John Wiley & Sons.

(DHLA) for biomedical devices [106]. The MPC units are responsible for anti-biofouling property while cross-linking can be induced by thiol-ene click chemistry between the thiol groups on DHLA and a trifunctional vinyl cross-linker for desired mechanical robustness

(Fig. 9). The as-prepared coatings showed no cytotoxicity to L929 cells and efficiently resisted the biofilm formation from *S. aureus* bacteria for up to 7 days. This coating potentially can be applied to tubular catheters or biomedical implants.

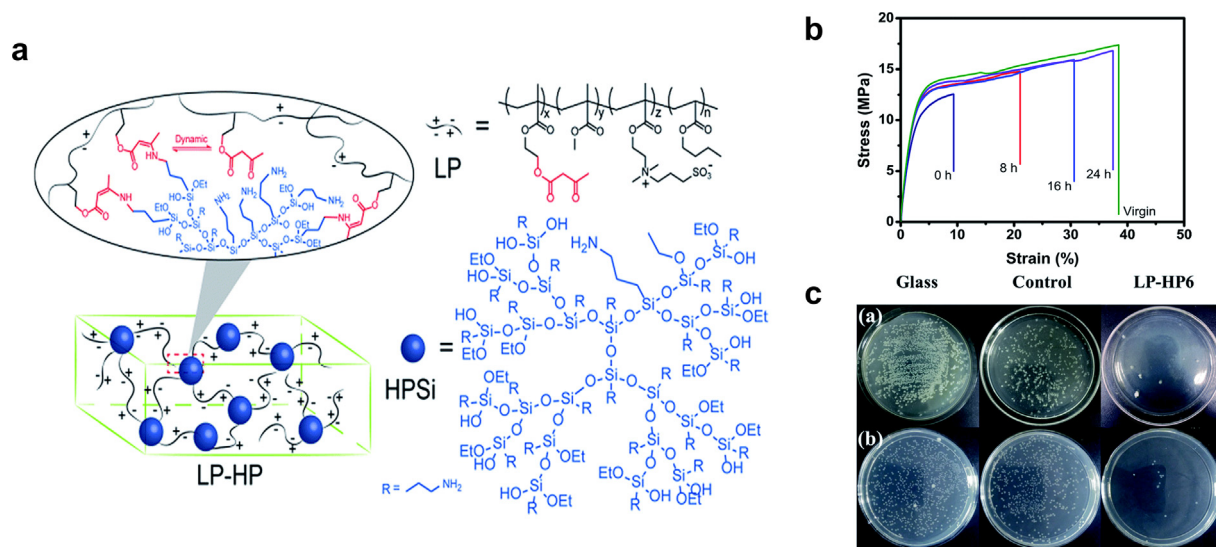


Fig. 10. (a) Chemical structures of the linear polyacrylate resin (LP), hyperbranched polysiloxane (HPSi), and LP-HP coatings. (b) Stress strain curves of the scratched coatings after healed for different length of time. (c) Antibacterial properties of glass, control, and LP-HP6 coatings toward (a) *E. coli* and (b) *S. aureus* colonization after for 3 h of contact. Reprinted with permission [105]. Copyright 2017, Royal Society of Chemistry.

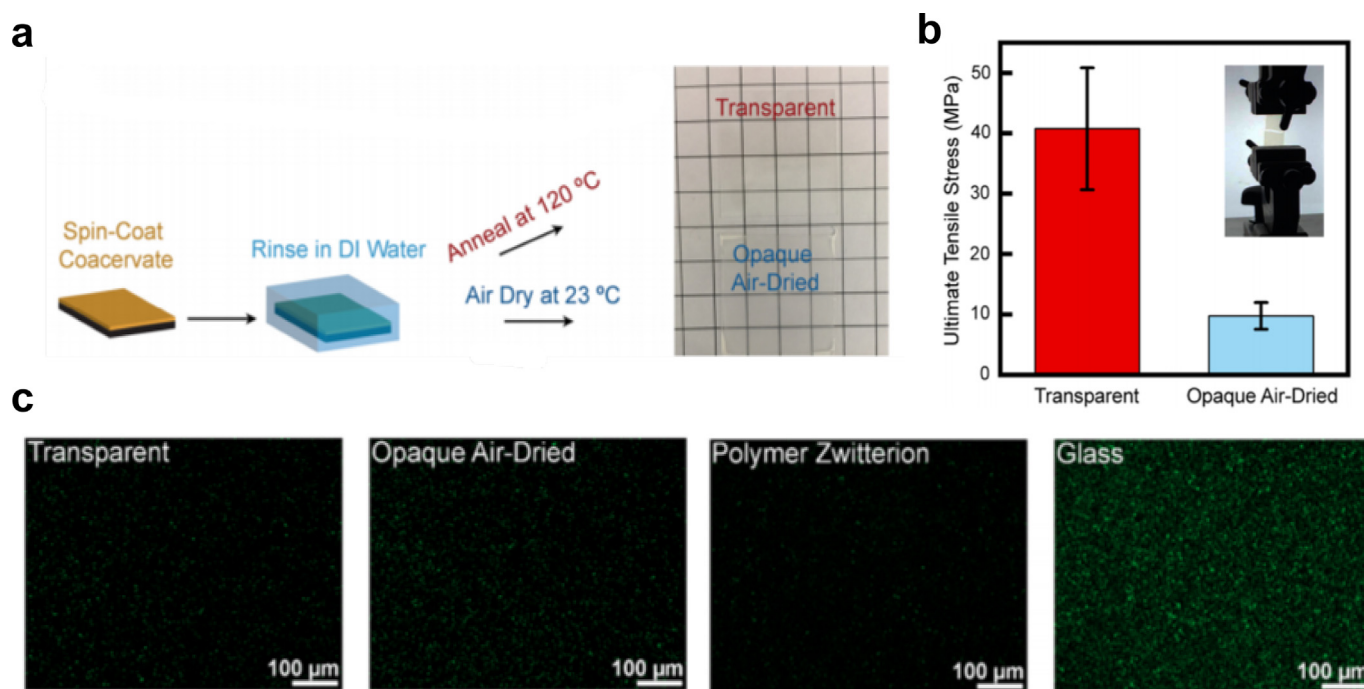


Fig. 11. (a) Fabrication illustration and photographs of the transparent and opaque air-dried PEC films. (b) Ultimate tensile stress of the PEC films. (c) Fluorescence microscopy images of the PEC films, polymer zwitterion, and glass surfaces after incubated with *E. coli* for 24 h. Reprinted with permission [106]. Copyright 2019, American Chemical Society.

Besides developing cross-linked coating structure via thiol-ene click chemistry, dynamic vinyllogous urethane linkage was also proven to be effective for obtaining networked structure with high mechanical strength. Gu et al. developed a multifunctional anti-biofouling coating with outstanding mechanical strength and reversible self-healing property based on this strategy [107]. Briefly, a linear polyacrylate resin (LP)

with sulfobetaine zwitterion and acetoacetyloxy groups was cross-linked with multi-amine hyperbranched polysiloxane (HPSi) to combine anti-biofouling and self-healing abilities (Fig. 10a). Fig. 10b displays the tensile strength test of the coatings with blade cut after different time length of self-healing. After healed for 24 h, the mechanical property of the coating was recovered to 94.07% of the original sample.

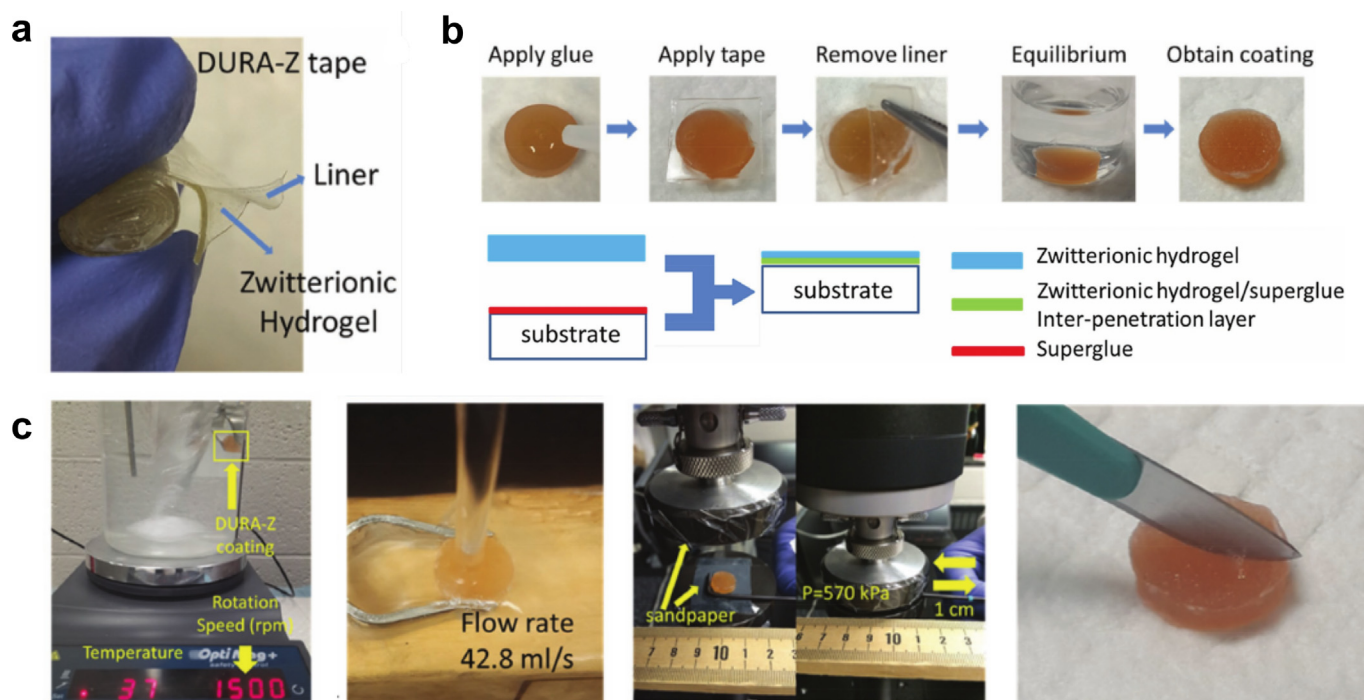


Fig. 12. (a) Structure of the DURA-Z tape. (b) Fabrication of DURA-Z coating on PU substrate and schematic illustration of the formation of DURA-Z coating. (c) Photographs of the durability test setup for 30 d exposure to PBS shearing, 30 d exposure to perpendicular water flush, 20 cycles of abrasion test under 570 kPa, and random scratch by a scalpel, respectively. Reprinted with permission [107]. Copyright 2017, John Wiley & Sons.

The high self-healing efficacy was attributed to the rich amine groups and sufficient chain segment motions of the HPSi. The antibacterial attachment efficacy was measured by incubating the adhered bacteria on the coating surfaces after 3 h exposure to bacterial suspension. Compared with bare glass and control samples, more than 95% of bacteria were prevented from attaching to the LP-HP6 coating (Fig. 10c).

In addition to constructing bacteria-repellent surfaces from zwitterionic polymers, polyelectrolyte complexes (PECs) consisting of oppositely charged polymers were also established as a straightforward way to prepare robust bacteria-resistant surfaces. In 2019, Schiffman et al. fabricated ultralow bacteria-fouling complex coacervates films based on anionic poly(4-styrenesulfonic acid) (PSS) and cationic poly(diallyldimethylammonium chloride) (PDADMAC) polymers [108]. The fabrication of the transparent and opaque air-dried PEC films was depicted in Fig. 11a, highly transparent films were obtained from thermal annealing due to the increase in the concentration of ionic bonds and the removal of stress-localizing defects. The ultimate tensile stress of the transparent is 4.2 times greater than that of the air-dried film which further confirmed this hypothesis (Fig. 11b). The anti-biofouling property of the PEC films was assessed by immersing in *E. coli* suspension for 24 h. A reduction of bacterial attachment by ~75% was observed from the PEC films compared to glass substrates (Fig. 11c).

Zwitterionic hydrogel as a widely investigated non-fouling material have been developed into durable anti-biofouling coating. Cao and colleagues presented a simple route to obtain a superdurable DURA-Z tape by in situ growing of a layer of poly(3-((3-acrylamidopropyl)-dimethylammonio)propanoate)) (PCBAA) zwitterionic hydrogel on polypropylene liner (Fig. 12a) [109]. This hydrogel tape can be readily binded to PU substrate using superglue, the strong adhesion of the DURA-Z tape to PU substrate was facilitated by the formation of zwitterionic hydrogel and superglue inter-penetration layer (Fig. 12b). The durability of the DURA-Z tape was evaluated under various challenging conditions, including 30 days of PBS buffer shearing, 30 days of exposure to perpendicular water flush, 20 cycles of abrasion test under 570 kPa, and random scratch by a scalpel (Fig. 12c). Moreover, the DURA-Z tape can also effectively prevent *E. coli* colonization with almost “zero” attachment for 30 days. This simple yet inexpensive technique demonstrated the feasibility of utilizing zwitterionic hydrogels for practical anti-biofouling applications.

5. Conclusions

In this review, we have presented the development strategies of durable anti-biofouling coatings for long-term applications, including the design concepts, underlying mechanisms, and methodologies to evaluate the anti-biofouling performance and robustness of these coatings. Besides the excellent long-lasting anti-biofouling properties obtained so far, most of the methods still show limited potential for practical application due to their complexity in fabrication as well as time and cost ineffectiveness. On the other hand, although these aforementioned anti-biofouling coatings have exhibited outstanding robustness in lab environment, most of the coatings still could not reach the criteria to cope with real world challenges. For instance, the abrasion resistance and self-healing properties of the coatings were only evaluated for limited cycles. The lack of assessment for more severe mechanical and chemical damage indicates that a myriad of these coatings still could fail over long-term service. Besides, the design criteria and durability test protocols of the durable anti-biofouling coatings should be adapted based on their applications. In summary, the design and fabrication of hydrophilic polymers-based coatings with integrated excellent anti-biofouling property, mechanical and chemical durability still have tremendous room to develop. We hope this review will provide insights for researchers to create new strategies hydrophilic polymer coatings with practical application in the future.

6. Future outlook

To enable the state-of-the-art hydrophilic polymers-based coatings for real world applications, on-site evaluations are inevitable. Specifically, coatings for biomedical implants/devices should be tested in vivo since platelet adhesion and blood coagulation phenomena could be more complicated than in vitro. Marine antifouling coatings are also supposed to be evaluated under both static and flow conditions in real maritime environment. Due to the diversity and complexity of the microorganisms and inorganic foulants, even the most well-simulated testing in lab still cannot guarantee the long-term performance of the coatings.

Since durable hydrophilic polymers-based anti-biofouling coatings often involves the incorporation of hydrophobic robustness reinforcement materials, the tradeoff between anti-biofouling potency and long-term durability can be well balanced by application-oriented testing standards. The structure-durability relationship should be scrutinized based on the application scenes to ensure the least reinforcing materials are introduced. Future research on increasing durability of these hydrophilic polymer-based anti-biofouling coatings should be directed toward discovering novel reinforcing strategies and configuring rigorous test protocols for persuasive demonstration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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